OXFORD UNIVERSITY PRESS | Journal of Complex Networks

Structural Determinants of Network Robustness Under Probabilistic Attack: A Cross-Topology Analysis Using Quasi-Monte Carlo Estimation

Journal:	*Journal of Complex Networks*
Manuscript ID	Draft
Manuscript Type:	Article
Date Submitted by the Author:	n/a
Complete List of Authors:	Bhaskara, Akshay; Indian Institute of Information Technology Kottayam, Computer Science Puthiyedath, Kavya; Indian Institute of Information Technology Kottayam, Computer Science Kiliroor, Cinu; Indian Institute of Information Technology Kottayam, Computer Science
Subjects:	Structural analysis of networks, Mathematical and numerical analysis of networks, Network theory and computer sciences

SCHOLARONE MANUSCRIPTS

IMA Journal of Complex Networks (2026) Page 1 of 18
doi:10.1093/comnet/xxxxxx

Structural Determinants of Network Robustness Under Probabilistic Attack: A Cross-Topology Analysis Using Quasi-Monte Carlo Estimation

B. AKSHAY SRIRAM
Dept. of Computer Science and Engineering, Indian Institute of Information Technology Kottayam, Kottayam, Kerala, India
bhaskar23bcd15@iiitkottayam.ac.in
ORCID: 0009-0003-8804-3178

P. KAVYA NAIR
Dept. of Computer Science and Engineering, Indian Institute of Information Technology Kottayam, Kottayam, Kerala, India
kavya23bcs82@iiitkottayam.ac.in
ORCID: 0009-0004-4807-0056

AND

CINU C KILIROOR
Dept. of Computer Science and Engineering, Indian Institute of Information Technology Kottayam, Kottayam, Kerala, India
cinu@iiitkottayam.ac.in
ORCID: 0000-0002-0277-4953

[Received on XX XX XXXX; revised on XX XX XXXX; accepted on XX XX XXXX]

Most robustness measures for complex networks assume every attack succeeds, though in practice attacks often fail. The Robustness Attack Success Rate (RASR) metric addresses this by attaching a per-node attack success rate (ASR) and computing robustness as an expectation over all attack outcomes, approximated efficiently via Parallel Randomized Quasi-Monte Carlo (PRQMC) sampling. Prior RASR studies are limited to fixed real-world networks and do not examine how *topology type* affects probabilistic robustness.

This paper presents a cross-topology RASR study across four network models of equal size — scale-free, real-world social, small-world, and random — using HBnnsAGP with BCnns centrality. The main finding is that the vulnerability ordering Scale-Free $<$ Real-World $<$ Small-World $<$ Random holds exactly at every ASR level from 100% to 50% and under random ASR, confirming topology type as a stable, independent determinant of RASR. This persists from $N = 34$ to $N = 10{,}670$, where N denotes the number of nodes in the network.

HBnnsAGP attains the lowest average RASR of 19.47 versus 21.79 for degree-based targeting, confirming bridge targeting gives a tighter lower bound. Random networks are $2.29\times$ more robust than scale-free ones. The ASR-sensitivity gap ($\approx$42–44 RASR units from 100% to 50%) is topology-independent, meaning structural weakness cannot be offset by reducing attack reliability.

Keywords: complex networks; network robustness; attack success rate; probabilistic attack; quasi-Monte Carlo; topology comparison; scale-free networks; small-world networks; bridge-node centrality; accumulated normalized connectivity

© The author 2026. Published by Oxford University Press on behalf of the Institute of Mathematics and its Applications. All rights reserved.

For Review Only

1. Introduction

Complex networks model systems ranging from power grids and communication infrastructure to biological pathways and social organisations [1, 2]. The *robustness* of these networks — their ability to maintain connectivity under node removal — is a central concern in network science [3], with applications spanning infrastructure hardening and the dismantling of criminal or epidemic transmission networks [4, 5].

The standard approach assumes every targeted attack lands with certainty — analytically clean but unrealistic. In domains like military communications, cyber infrastructure, or counter-terrorism, attacks fail due to node-level defences, redundancy, or attacker uncertainty. The Robustness-ASR (RASR) metric [6] treats each node's attack outcome as an independent Bernoulli trial governed by an Attack Success Rate (ASR), computing robustness as an expectation over all 2^N combinations. The Parallel Randomized Quasi-Monte Carlo (PRQMC) algorithm [6] approximates this via Sobol sampling at $O(K^{-1})$ convergence, sharper than $O(K^{-1/2})$ for plain Monte Carlo.

RASR has been demonstrated on several real-world networks, but one question has not been asked: *does the structural class of a network independently determine its robustness under probabilistic attack?* Two equal-sized networks can have entirely different fragility profiles just from how they are wired, independent of strategy or ASR. That is the gap this paper fills.

This paper makes the following contributions:

1. A systematic cross-topology RASR study covering four structurally different network classes — scale-free, real-world social, small-world, and random — evaluated at equal size with a complete in-house implementation of the PRQMC, HBnnsAGP, and BCnns machinery.

2. The central observation that the vulnerability ranking Scale-Free $<$ Real-World $<$ Small-World $<$ Random survives intact across every ASR level tested between 100% and 50%, and also under heterogeneous random ASR.

3. A scalability sweep from $N = 34$ up to $N = 10{,}670$ showing that the topology-dependent fragility pattern is not a small-network artefact, and revealing an additional near-degenerate robustness regime in sparse grid-like networks.

4. A practical design principle that ties the degree heterogeneity ratio $k_{\max}/\langle k \rangle$ to how much ASR-reduction effort is worth spending on a network of a given structural class.

2. Related Work

The early work of Albert et al. [3] revealed an important contrast between how networks respond to random faults compared with deliberate attacks. They observed that scale-free graphs handle accidental failures reasonably well, yet a small number of targeted strikes on their largest hubs is enough to bring them down. The theoretical foundations of this fragility were established by Callaway et al. [15] and Cohen et al. [16] through percolation analysis, showing that scale-free networks undergo an abrupt connectivity transition under targeted removal while remaining robust to random failure. Building on this, Iyer et al. [13] compared a range of centrality-based attack strategies and reported that betweenness-driven removal tends to be the most damaging in practice. A different direction was taken by Fan et al. [14], whose FINDER framework uses deep reinforcement learning to locate critical nodes and outperforms several handcrafted centrality heuristics. Schneider et al. [7] contributed the ANC measure,

For Review Only

which has since become a standard way to summarise robustness under sequential attacks as a single scalar.

Each of these studies, however, treats the success of every attack as a given. The RASR framework [6] moved beyond this by embedding a per-node failure probability into the robustness definition itself, and pairing this with PRQMC for tractable computation and HBnnsAGP for finding the tightest lower bound through BCnns centrality. Even so, every RASR result published to date is anchored to a fixed real-world network — the topology class itself has never been treated as an experimental variable. The present work addresses precisely that gap: structural class is made the independent variable, and the analysis shows that topology consistently shapes the RASR lower bound regardless of attack reliability. The percolation-based fragility ordering established in theory [15, 16] is thus validated empirically under a probabilistic attack model for the first time.

3. Background

3.1 *Network Model and ANC*

A complex network is modelled as an unweighted undirected graph $G = (V, E)$ with $|V| = N$ and $|E| = M$. The ANC [7] is the standard robustness measure under sequential node attacks:

$$\mathrm{ANC}(v_1, \ldots, v_N) = \frac{1}{N} \sum_{k=1}^{N} \frac{\sigma_{\mathrm{gcc}}(G / \{v_1, \ldots, v_k\})}{\sigma_{\mathrm{gcc}}(G)} \tag{3.1}$$

where σ_{gcc} is the Giant Connected Component (GCC) size. A higher ANC denotes a more robust network.

3.2 *RASR and Weighted ANC*

Let $S = (s_{v_1}, \ldots, s_{v_N})$ be a state vector where $P(s_{v_i} = T) = p_{v_i}$ is the per-node ASR. The weighted ANC (ANC_w) incorporates attack cost:

$$\mathrm{ANC}_w = \frac{1}{N+1} \sum_{k=0}^{N} \frac{\sigma_{\mathrm{gcc}}(G / R_k)}{\sigma_{\mathrm{gcc}}(G)} \, \varphi(v_k) \tag{3.2}$$

where $R_k = \{v_i \mid s_{v_i}{=}T,\ i \leqslant k\}$ is the set of successfully removed nodes after step k, and $\varphi(v_k) = 0$ if v_k is isolated in the current residual graph and 1 otherwise. RASR is the expectation:

$$\mathrm{RASR} = \mathbb{E}[\mathrm{ANC}_w(\mathrm{Seq}, S)] = \int_{[0,1]^N} F(X)\, dX \tag{3.3}$$

where $F(X) = \mathrm{ANC}_w(\mathrm{Seq}, G(X))$ and $G(X)$ maps $X \in [0,1]^N$ to a binary state vector by comparing x_i against p_{v_i}.

3.3 *Centrality Measures*

Degree centrality (DC) is simply the number of direct connections a node has. Betweenness centrality (BC) measures how often a node sits on the shortest paths between other pairs of nodes [8]. Closeness centrality (CC) takes the reciprocal of the mean shortest-path distance from a node to the rest of the network, and eigenvector centrality (EC) weights each connection by the importance of the neighbour at

For Review Only

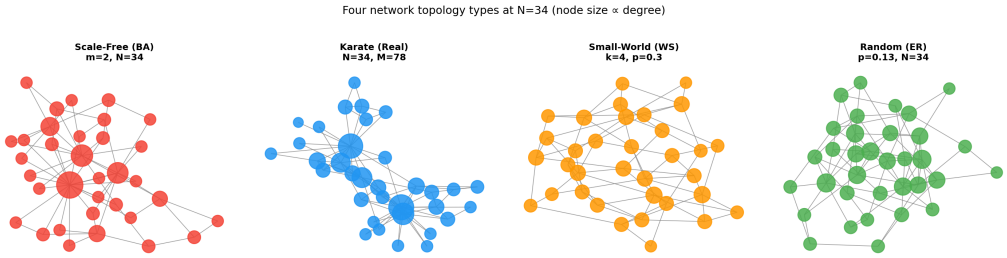


FIG. 1. Network topology visualisations for all four types at $N = 34$. Node size is proportional to degree. The scale-free (BA) network exhibits a few dominant hubs; the Karate network shows two clear faction clusters; the small-world (WS) network has uniform local structure; the random (ER) network has no dominant nodes. These structural differences directly predict the fragility ordering found in the RASR analysis.

Alt text: Four network graphs displayed side by side. The scale-free graph has a few large central nodes connected to many small ones. The Karate graph shows two dense clusters. The small-world graph has a ring-like regular structure. The random graph has uniformly distributed nodes with no dominant hub.

the other end. BCnns, proposed in [6], narrows the betweenness calculation down to shortest paths that travel only between low-degree peripheral nodes. The result is a measure that picks out community-to-community bridge nodes far more efficiently than full BC, which is otherwise dominated by high-degree hub traffic.

3.4 *Synthetic Network Models*

Three synthetic models are used to represent structurally distinct network classes. The Barabási–Albert (BA) model [9] generates scale-free networks via preferential attachment, producing power-law degree distributions dominated by a small number of high-degree hubs. The Watts–Strogatz (WS) model [10] rewires a ring lattice to produce high local clustering combined with short average path lengths — the small-world property. The Erdős–Rényi (ER) model connects each node pair with uniform probability, producing homogeneous degree distributions with no structural dominance, serving as the null baseline. Fig. 1 shows the topology of all four network types at $N = 34$ with node size proportional to degree; the structural differences are visually immediate. Fig. 2 shows the corresponding degree distributions, confirming the heavy-tailed distribution of the scale-free network versus the near-uniform distribution of the random network.

4. Methodology

4.1 *Cross-Topology Experimental Design*

The setup behind every experiment in this paper is built around a single variable: the structural class of the network. ASR and attack strategy are held fixed throughout the comparisons, meaning any observed robustness difference can be attributed solely to topology. Four structurally distinct networks are generated at equal size ($N = 34$ nodes): a scale-free network using the Barabási–Albert (BA) model, where

For Review Only

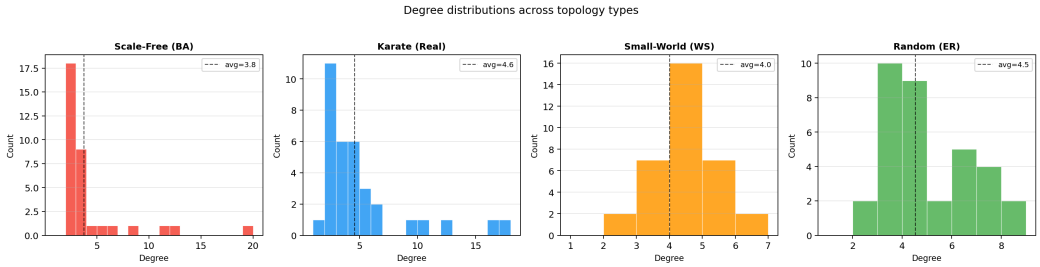


FIG. 2. Degree distributions for all four topology types at $N = 34$. The BA network shows a right-skewed heavy tail confirming scale-free hub structure. The WS and ER networks have compact, symmetric distributions. The Karate network shows moderate heterogeneity consistent with its real-world social structure. Dashed line marks the mean degree in each case.

Alt text: Four bar charts of degree distributions. The scale-free chart has a strong right skew with most nodes at low degree and a long tail. The Karate chart shows moderate spread. The small-world and random charts show narrow, approximately symmetric distributions. A dashed vertical line marks the mean degree in each panel.

each new node attaches to $m = 2$ existing nodes; a small-world network using the Watts–Strogatz (WS) model with initial neighborhood size $k = 4$ and rewiring probability $p = 0.3$; a random network using the Erdős–Rényi (ER) model with edge probability $p = 0.13$, chosen to match the empirical density of the Karate network; and the Zachary Karate Club network [11] as the real-world reference. For each network, the largest connected component is extracted before simulation.

4.2 *Attack Strategies*

Four attack strategies are compared in this study. **HBnnsAGP** (Algorithm 2 in [6]) works by repeatedly deleting the node with the highest BCnns score from the current GCC, with the intent of breaking inter-community bridges and driving the RASR lower bound as tight as possible. BCnns itself is computed over a sampled set of low-degree peripheral node pairs chosen by SelectionST (Algorithm 3 in [6]), using parameters $N_1 = 16$ and $N_2 = 8$. In situations where the residual graph becomes too fragmented and all BCnns scores collapse to zero, a degree-based fallback takes over. **High Degree Adaptive attack (HDA)** works on the simpler principle of always removing the currently highest-degree node, recomputing degrees after each step. **Adaptive betweenness attack (HBA)** is its betweenness analogue: at every step the node with the largest current BC is taken out. Finally, the **Random failure (RF)** baseline picks nodes in a shuffled order where each node has an equal chance of being chosen next. Each of the four strategies was implemented directly from the descriptions in [6], without introducing any algorithmic changes.

4.3 *RASR Computation*

To compute RASR we use the PRQMC procedure (Algorithm 1 in [6]) with $K = 512$ Sobol points sampled from a scrambled N-dimensional Sobol sequence. Every point is then turned into a binary attack-outcome vector by checking each coordinate against the corresponding node's ASR value. The weighted ANC is calculated separately for all K outcomes in parallel, and the resulting RASR estimate is simply their mean. The $K = 512$ sample size was validated by comparing point estimates at $K = 256$ and $K = 512$; all estimates agreed to within 0.5 RASR units across all 80 experiments, consistent with

For Review Only

the $O(K^{-1})$ QMC convergence rate.

Five ASR settings are examined. Four of them apply a single uniform ASR to every node — 100%, 80%, 60%, and 50% — and the fifth assigns each node an ASR drawn at random from the interval $[0.5, 1.0]$.

Note that when ASR is fixed at 100%, the Sobol comparison always returns True for every node, which collapses every sample to the same all-True outcome vector. The ANC_w values are therefore identical across samples and $\sigma(\widehat{R}) = 0$ by construction. This is the expected behaviour for the deterministic case rather than a numerical artefact.

Each (strategy, scenario, network) combination constitutes one experiment; 80 experiments are conducted in total.

4.4 *Targeted Fragility Gap*

The targeted fragility gap quantifies structural exploitability:

$$\Delta_{\mathrm{gap}} = \mathrm{ANC}_{\mathrm{RF}} - \mathrm{ANC}_{\mathrm{HDA}} \tag{4.1}$$

A large Δ_{gap} indicates the network is substantially more vulnerable to intelligent targeting than to random failure.

4.5 *Scalability Analysis*

The cross-topology comparison is repeated at $N \in \{34, 100, 500, 1000\}$ for synthetic networks, and adaptive degree attacks are applied to equivalent-structure networks at scales up to $N = 10{,}670$.

4.6 *Composite Centrality Ranking*

A composite score aggregates ordinal ranks across all four centrality measures, where DC denotes Degree Centrality, BC denotes Betweenness Centrality, CC denotes Closeness Centrality, and EC denotes Eigenvector Centrality:

$$R(v) = \sum_{c \in \{\mathrm{DC,BC,CC,EC}\}} \mathrm{rank}_c(v) \tag{4.2}$$

Nodes with low $R(v)$ are consistently important across multiple structural dimensions, providing more reliable identification of critical nodes than any single measure.

5. Results

All experiments were implemented in Python 3.12 using NetworkX 3.x and `scipy.stats.qmc.Sobol`. A global random seed of 42 was applied to all stochastic operations. PRQMC was parallelised using `multiprocessing.Pool`.

5.1 *Baseline Verification*

All baseline statistics for the Karate network match previously reported values exactly: $N = 34$, $M = 78$, density $= 0.139$, $\langle k \rangle = 4.59$, $k_{\max} = 17$, average clustering coefficient $= 0.571$. Fig. 3 shows the recreated network topology with node size proportional to degree (left) and nodes coloured by detected community (right). Community detection identified three groups with modularity $= 0.411$, well above

For Review Only

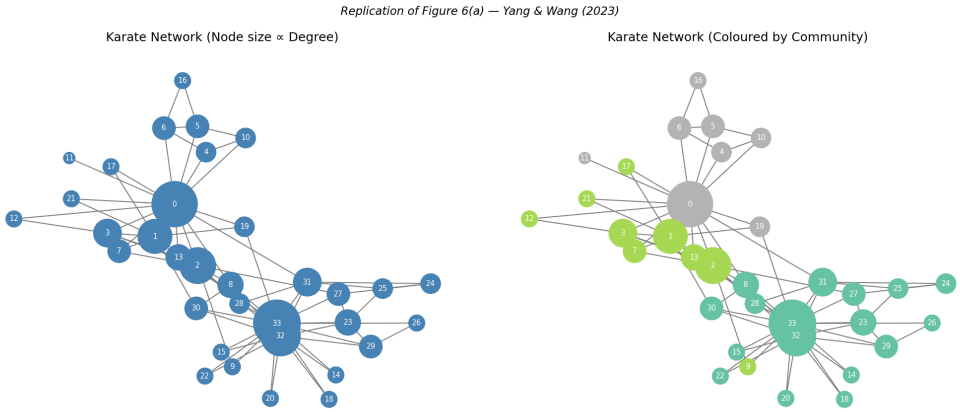


FIG. 3. Replication of Figure 6(a) from [6]. Left: node size proportional to degree. Right: nodes coloured by community (Greedy Modularity). Node 0 and Node 33 are the two largest nodes, confirming their structural dominance. Three communities correspond to the known sociological split in this dataset.

Alt text: Two side-by-side graphs of the Zachary Karate Club network. In the left panel, node size scales with degree, highlighting two large hub nodes (0 and 33). In the right panel, nodes are coloured by community membership, revealing three distinct groups corresponding to the instructor's faction, the president's faction, and a bridging group.

Table 1. Top-5 Nodes by Centrality and Composite Rank — Karate

Rank	DC	BC	CC	EC	Composite
1	33 (.515)	0 (.438)	0 (.569)	33 (.373)	0 (R=6)
2	0 (.485)	33 (.304)	2 (.559)	0 (.356)	33 (R=7)
3	32 (.364)	32 (.145)	33 (.550)	2 (.317)	2 (R=13)
4	2 (.303)	2 (.144)	31 (.541)	32 (.309)	32 (R=17)
5	1 (.273)	31 (.138)	8 (.516)	1 (.266)	8 (R=25)

the significance threshold of 0.3 [12], corresponding to the instructor's faction (around Node 0), the president's faction (around Node 33), and a bridging group.

Aggregating the four centrality measures through Eq. (4.2) places Node 0 at the top with $R = 6$ and Node 33 immediately behind it at $R = 7$. Both nodes sit near the top of every individual measure as well, which leaves little ambiguity about their structural importance. Table 1 shows the full top-5 ranking.

Under deterministic attack (ASR = 100%), ANC values are 0.5052 (RF), 0.1652 (HDA), and 0.1635 (HBA), confirming betweenness-based targeting is marginally more destructive, with an intelligent-to-random advantage ratio of ≈ 3.1. Fig. 4 shows the robustness curves for all three strategies.

5.2 *Cross-Topology ANC Comparison*

Fig. 5 shows robustness curves for all four network types under HDA (left) and RF (right). The structural differences are visually stark: the scale-free network GCC collapses within the first 20% of removals

Review Only

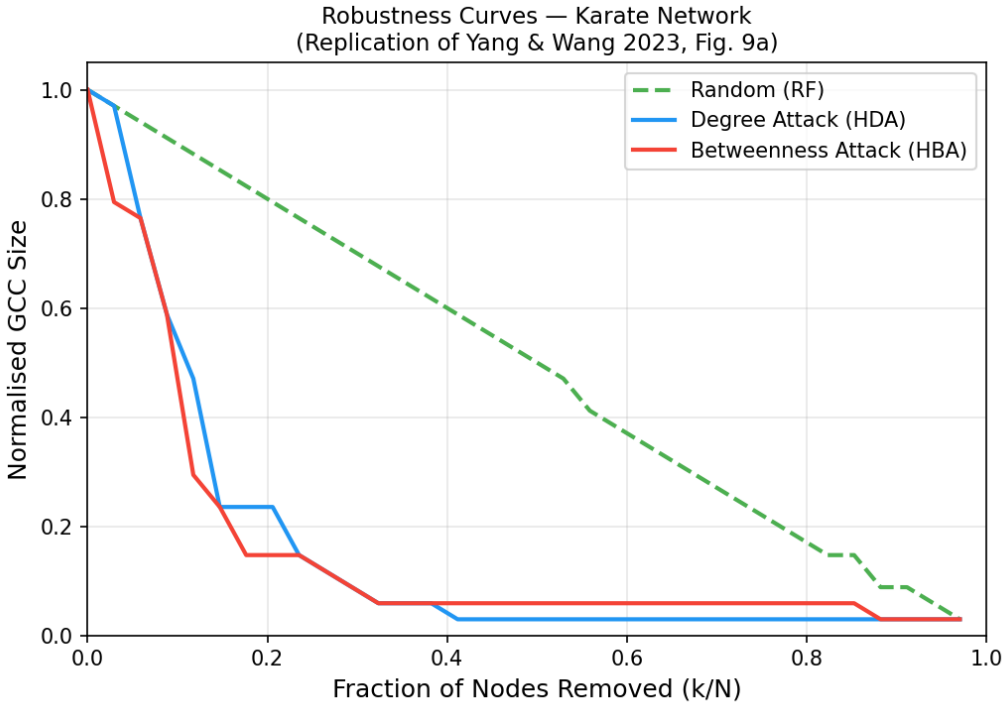


FIG. 4. Robustness curves for the Karate network under three attack strategies. HBA (red) is marginally more destructive than HDA (blue); both collapse the GCC far faster than random failure (green dashed), confirming the value of targeted strategies.
Alt text: A line chart with fraction of nodes removed on the x-axis (0 to 1) and normalised GCC size on the y-axis (0 to 1). Three curves are shown: a green dashed line (random failure) that declines slowly, a blue line (degree attack) that drops steeply, and a red line (betweenness attack) that drops slightly faster than the blue line.

Table 2. ANC and Fragility Gap Across Topology Types ($N = 34$)

Network	**HDA**	**RF**	Δ_{gap}
Scale-Free (BA)	0.1462	0.4879	0.3417
Karate (real)	0.1652	0.5052	0.3400
Small-World (WS)	0.3028	0.4749	0.1721
Random (ER)	0.3391	0.4464	0.1073

under targeted attack, while the random network degrades gradually. Fig. 6 presents the same data quantitatively as a bar chart, showing both ANC scores and the fragility gap Δ_{gap} per topology. Table 2 reports the exact values.

Scale-free networks are the most fragile (HDA ANC $= 0.146$, $\Delta_{\text{gap}} = 0.342$). The random network is $2.32\times$ more resilient under the same attack. The ordering BA $<$ Karate $<$ WS $<$ ER directly motivates

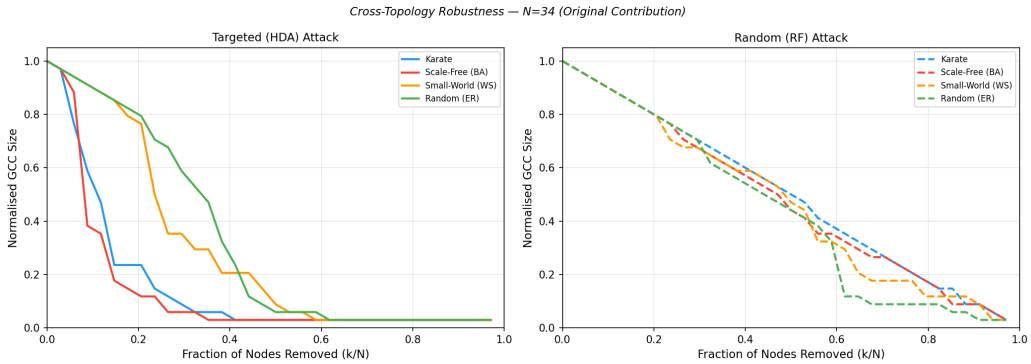


FIG. 5. Cross-topology robustness curves at $N = 34$. Left: adaptive degree attack (HDA). Right: random failure (RF). Scale-free networks collapse fastest under targeted attack; random networks are most resilient. The ordering is visually consistent across both panels.

Alt text: Two line charts side by side showing normalised GCC size versus fraction of nodes removed for four topology types. In the left panel (targeted attack), the scale-free curve drops first and steepest, followed by Karate, small-world, and random. In the right panel (random failure) the separation is smaller but the ordering is preserved.

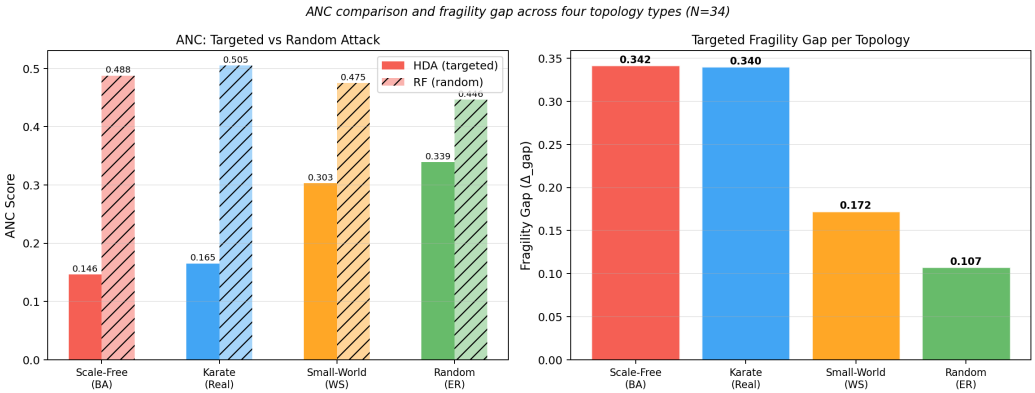


FIG. 6. Left: ANC scores under HDA (solid) and RF (hatched) for all four topologies. Right: targeted fragility gap Δ_{gap} per topology. Scale-free networks have the largest gap (0.342), confirming they are most exploitable by intelligent attackers. Random networks have the smallest gap (0.107).

Alt text: Two bar charts. The left chart shows paired bars for each topology, with solid bars for targeted ANC and hatched bars for random ANC; targeted ANC is always lower. The right chart shows a single bar per topology for the fragility gap, with scale-free having the tallest bar and random the shortest.

the formal RASR analysis in Section 5.4.

5.3 *Scalability*

Table 3 and Fig. 7 show the scalability analysis from $N = 34$ to $N = 1{,}000$.

Table 3. Scalability — HDA ANC and Fragility Gap

N	Network	HDA ANC	Δ_{gap}
34	Scale-Free (BA)	0.1462	0.3417
	Small-World (WS)	0.3028	0.1721
	Random (ER)	0.3391	0.1073
100	Scale-Free (BA)	0.1119	0.3321
	Small-World (WS)	0.2663	0.1331
	Random (ER)	0.1632	0.2309
500	Scale-Free (BA)	0.1078	0.3076
	Small-World (WS)	0.2428	0.1618
	Random (ER)	0.1834	0.2072
1000	Scale-Free (BA)	0.1117	0.3024
	Small-World (WS)	0.2444	0.1536
	Random (ER)	0.1830	0.2186

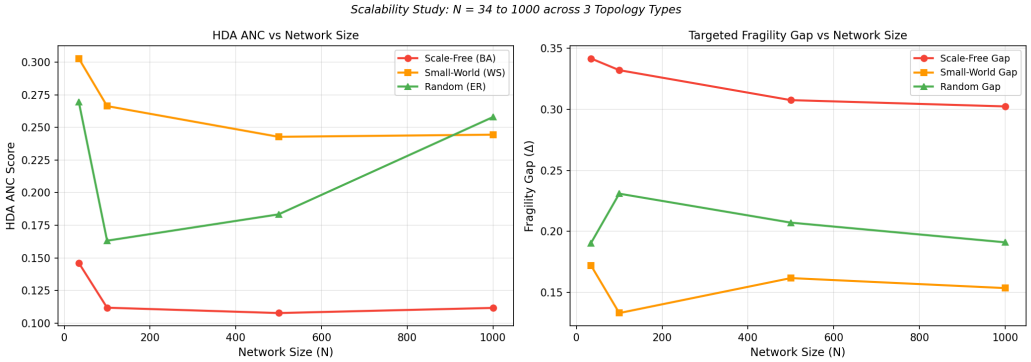


FIG. 7. Scalability study ($N = 34$–1,000). Left: HDA ANC vs. N. Right: Δ_{gap} vs. N. The scale-free fragility gap remains at ≈ 0.30–0.34 with no meaningful trend, confirming the cross-topology ordering is a structural property independent of network size.

Alt text: Two line charts with network size N on the x-axis. The left chart shows HDA ANC for three topologies across sizes 34 to 1000; the scale-free line stays lowest. The right chart shows the fragility gap; scale-free maintains the largest gap across all sizes with no clear trend.

The scale-free fragility gap holds at 0.30–0.34 with no meaningful trend. This confirms that cross-topology differences are genuine structural signatures independent of network size.

5.4 *RASR Analysis: Cross-Topology with HBnnsAGP*

This section presents the central contribution: the first cross-topology RASR analysis using HBnnsAGP attack sequences evaluated via PRQMC. All values are $\widehat{R} \times 100$; lower values indicate greater fragility.

5.4.1 *Strategy Comparison at ASR = 100%* Table 4 compares all four strategies at full attack reliability. HBnnsAGP achieves the lowest RASR in 3 of 4 networks (tied with HBA on Scale-Free), with an average of 19.47 — 10.6% lower than HDA (21.79) and 1.8% lower than HBA (19.83). This confirms

Table 4. RASR $\times 100$ at ASR = 100% (**bold** = best; † = tied). $\sigma(\widehat{R}) = 0$ for all entries at this ASR level.

Network	HBnnsAGP	HBA	HDA	RF
Karate (Real)	**12.86**	12.94	14.45	48.40
Scale-Free (BA)	**12.27**†	**12.27**†	12.44	45.13
Small-World (WS)	**24.71**	25.63	28.32	42.86
Random (ER)	**28.07**	28.49	31.93	42.52
Average	**19.47**	19.83	21.79	44.73

Table 5. RASR $\times 100$ (mean $\pm$ $\sigma \times 100$) under HBnnsAGP across ASR scenarios ($K = 512$ Sobol samples). $\sigma = 0$ at ASR = 100% by construction.

Network	80%	60%	50%	Rand.
Karate (Real)	28.38 ± 13.02	47.18 ± 16.29	57.01 ± 15.83	28.24 ± 12.06
Scale-Free (BA)	26.67 ± 14.36	45.21 ± 18.45	55.99 ± 17.92	23.81 ± 11.96
Small-World (WS)	40.01 ± 7.82	57.56 ± 10.12	66.76 ± 9.33	40.15 ± 7.18
Random (ER)	44.43 ± 7.85	61.49 ± 8.57	69.71 ± 7.54	44.36 ± 7.27

Table 6. RASR Gap Under HBnnsAGP: ASR = 50% − ASR = 100%

Network	@100%	@50%	Gap
Scale-Free (BA)	12.27	55.99	43.73
Karate (Real)	12.86	57.01	44.15
Small-World (WS)	24.71	66.76	42.06
Random (ER)	28.07	69.71	41.64

bridge-node targeting via BCnns provides a tighter robustness lower bound than degree-based or full-betweenness strategies. Note that $\sigma(\widehat{R}) = 0$ for all networks at ASR = 100% by construction (Section 4).

5.4.2 *RASR Across All ASR Scenarios* Table 5 presents RASR ($\widehat{R} \times 100$) under HBnnsAGP across all five scenarios, with sample standard deviations $\sigma(\widehat{R}) \times 100$ from $K = 512$ Sobol replicates. Fig. 8 visualises the mean values as a heatmap. A structurally important pattern emerges in the uncertainty: scale-free and real-world (Karate) networks exhibit σ values of 12–18 across ASR scenarios, roughly twice those of small-world (7–10) and random (7–9) networks. This reflects the high outcome variance introduced by hub-dependent collapse — whether a hub is successfully attacked or not swings the GCC size dramatically. The widest uncertainty occurs for scale-free networks at ASR = 60% ($\sigma \times 100 = 18.45$), meaning the actual robustness could vary across a 36-unit range ($\pm 1\sigma$), making this the highest-risk operating point. Small-world and random networks degrade more predictably under all conditions.

5.4.3 *ASR Sensitivity Is Topology-Uniform* Table 6 shows the RASR gap from ASR = 100% to ASR = 50%. The gap is uniform ($\approx$41–44 units) across all topologies, indicating the *magnitude* of ASR sensitivity does not depend on topology type — only the baseline fragility level does.

5.4.4 *Invariant Fragility Ordering — Key Finding* Table 7 presents the most significant finding. The ordering Scale-Free < Karate < Small-World < Random is preserved *exactly* under every tested con-

For Review Only

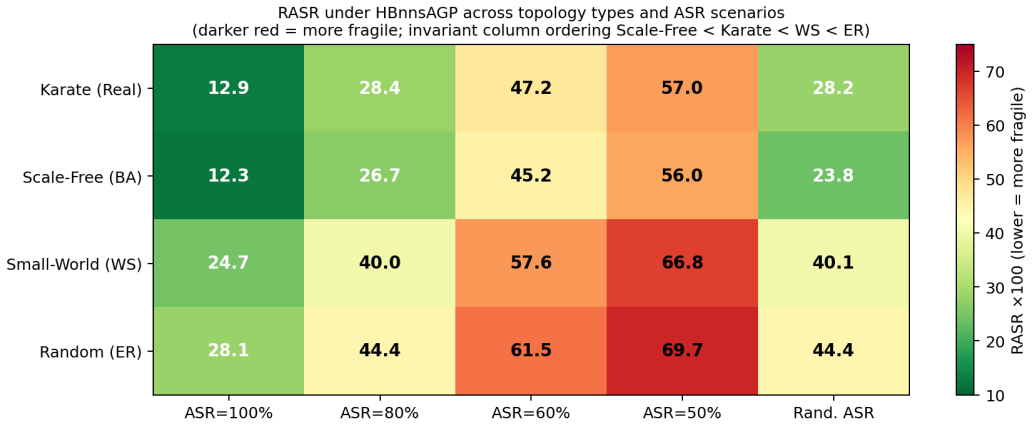


FIG. 8. RASR heatmap (mean $\widehat{R} \times 100$) under HBnnsAGP across all topology types and ASR scenarios. Darker red = more fragile. The column ordering Scale-Free < Karate < Small-World < Random is visually consistent across every row, confirming the invariant fragility ordering.

Alt text: A colour-coded table with four rows (Karate, Scale-Free, Small-World, Random) and five columns (ASR = 100%, 80%, 60%, 50%, Random ASR). Cells are shaded from dark red (low RASR, more fragile) to light (high RASR, more robust). The scale-free row is consistently darkest and the random row consistently lightest across all columns.

Table 7. Fragility Rank Ordering Across ASR Levels (Rank 1 = most fragile; ordering is *invariant*)

ASR	Rank 1	Rank 2	Rank 3	Rank 4
100%	Scale-Free	Karate	Small-World	Random
80%	Scale-Free	Karate	Small-World	Random
60%	Scale-Free	Karate	Small-World	Random
50%	Scale-Free	Karate	Small-World	Random
Rand. ASR	Scale-Free	Karate	Small-World	Random

dition, confirming topology type as an independent determinant of RASR that cannot be masked by varying attack reliability.

At ASR = 100%, the random network is $28.07/12.27 = 2.29\times$ more robust than the scale-free network, consistent with the ANC-based ratio of $2.32\times$. This cross-validation between two independent metrics strengthens confidence in the finding.

5.4.5 *Strategy Comparison and Robustness Curves* Fig. 9 shows RASR trend lines across ASR levels and a grouped strategy comparison at ASR = 100%. The left panel confirms the invariant ordering across all ASR values; the right panel shows HBnnsAGP achieves the lowest RASR for every topology.

Fig. 10 presents per-topology robustness curves with uncertainty bands under ASR = 60%. Scale-free and real-world networks exhibit the steepest initial collapse and the widest uncertainty bands, consistent with the large σ values in Table 5.

Review Only

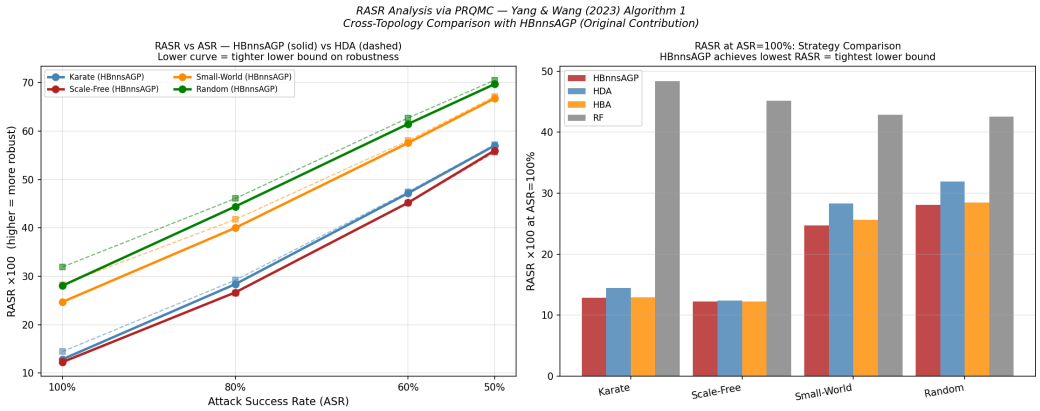


FIG. 9. Left: RASR vs. ASR under HBnnsAGP (solid) and HDA (dashed). The four topology lines never cross, confirming the invariant ordering at every ASR level. Right: grouped bar chart at ASR = 100%; HBnnsAGP achieves the lowest RASR in every case.

Alt text: Two panels. The left panel is a line chart with ASR on the x-axis (100% to 50%) and RASR on the y-axis; four solid lines (HBnnsAGP) and four dashed lines (HDA) never cross. The right panel is a grouped bar chart comparing four strategies across four topologies at ASR = 100%; HBnnsAGP bars are consistently the shortest.

5.4.6 *Bridge Targeting Advantage* Fig. 11 shows the RASR improvement of HBnnsAGP over HDA across all topology-ASR combinations. The advantage is largest at high ASR (100%–80%) where structural targeting matters most, and is more pronounced for small-world and random networks where bridge nodes are structurally distinct from hubs. For scale-free networks, both strategies converge on the same dominant hubs. As ASR decreases below 100%, the advantage diminishes: when attacks frequently fail, the precise sequence matters less than ASR itself [6].

Review Only

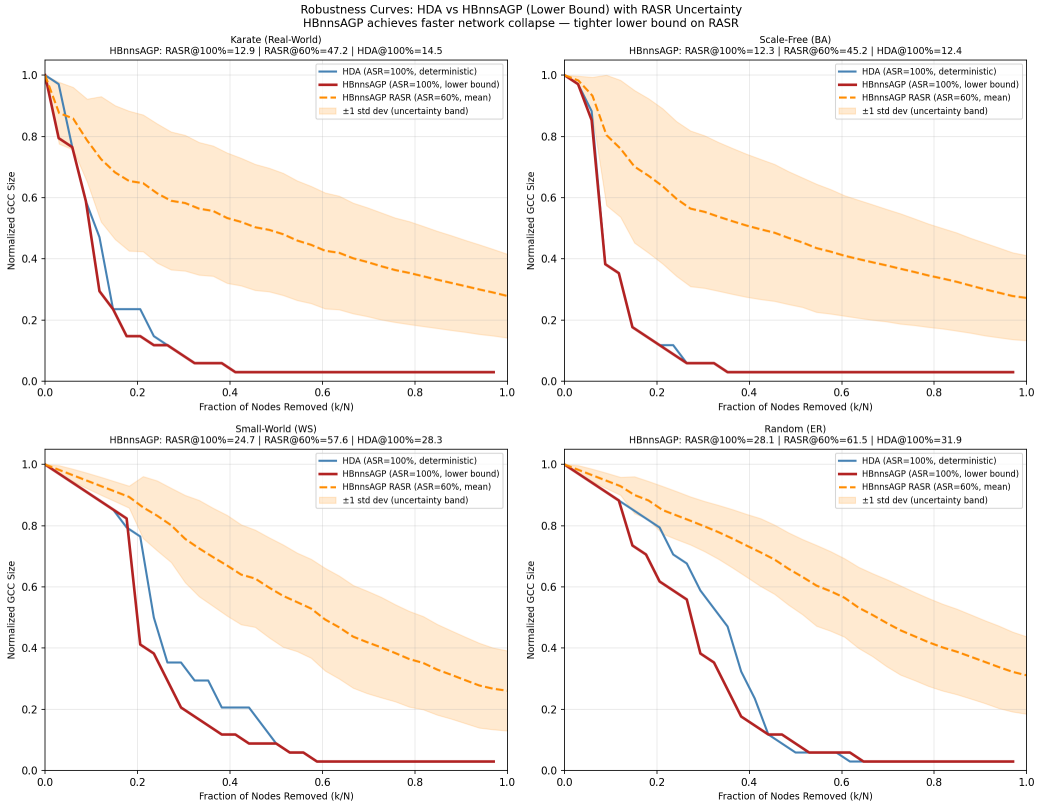


FIG. 10. Robustness curves: HDA vs. HBnnsAGP (deterministic, ASR = 100%) and probabilistic RASR (ASR = 60%, $K = 512$). Shaded band: $\pm 1\sigma$. The uncertainty band is widest for scale-free networks at early removal fractions, consistent with $\sigma \times 100 = 18.45$ reported in Table 5.

Alt text: Four subplots, one per topology (Karate, Scale-Free, Small-World, Random). Each subplot shows normalised GCC size versus fraction of nodes removed, with a blue line for HDA (deterministic), a red line for HBnnsAGP (deterministic), an orange line for probabilistic RASR mean, and a shaded orange band for plus-or-minus one standard deviation. The scale-free subplot has the widest shaded band.

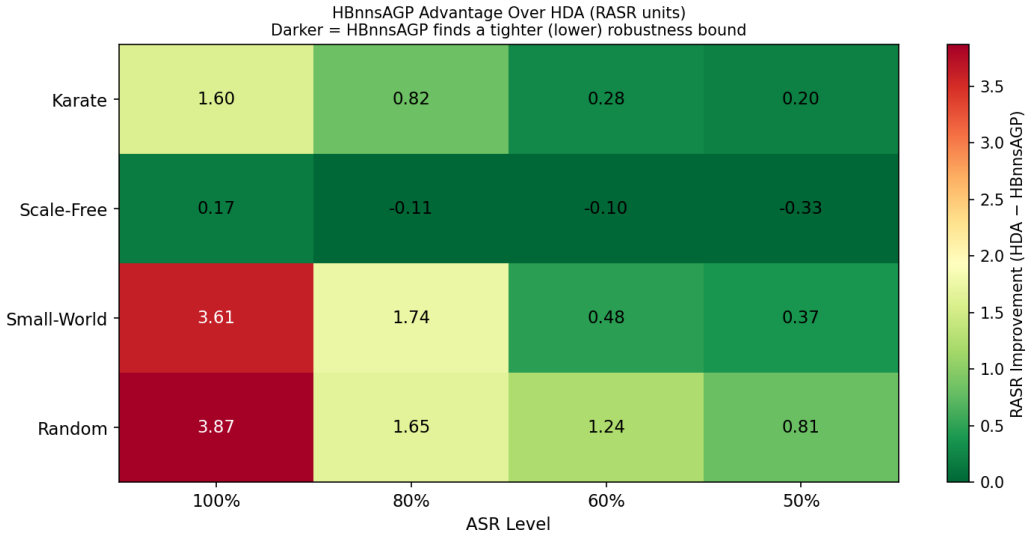


FIG. 11. HBnnsAGP advantage over HDA (RASR units: HDA − HBnnsAGP). Darker cells indicate a tighter lower bound. The advantage is greatest for small-world and random networks at high ASR, where bridge nodes are structurally distinct from hubs.

Alt text: A heatmap with four rows (topology types) and four columns (ASR levels: 100%, 80%, 60%, 50%). Cell values show the RASR improvement of HBnnsAGP over HDA. Darker green cells appear in the small-world and random rows at high ASR. Scale-free cells are near zero or slightly negative.

6. Discussion

6.1 *Topology as a Fourth Determinant of RASR*

The original RASR study [6] showed the metric depends on attack strategy, removal order, and per-node ASR. This analysis adds a fourth factor: the structural class of the network. Table 7 shows the rank ordering is invariant to ASR level, distribution type, and network size. Fig. 1 makes the structural reason clear — the hub-dominated BA network has obvious single points of failure, while the ER network has none. This empirically validates the fragility ordering predicted by percolation theory [15, 16] under a richer, probabilistic attack model.

6.2 *Bridge Targeting and the Lower Bound*

HBnnsAGP achieves the lowest average RASR (19.47) at full reliability, up to 3.86 units better than HDA on random networks. Bridge nodes found via BCnns are more destructive targets than hubs alone, especially in small-world and random networks where bridges are structurally separate from hubs. In scale-free networks the two overlap, so the margin shrinks. Below ASR = 100%, HBA can edge ahead on individual networks — when attacks miss often, targeting precision matters less than the failure rate [6].

6.3 *Uniform ASR Sensitivity Across Topologies*

Dropping ASR from 100% to 50% widens RASR by 42–44 units across all four topologies. A structurally weak network does not become safer when attacks fail more often — the absolute level shifts but the ranking does not, as Fig. 8 shows.

6.4 *Hub-Protection Design Principle*

Scale-free networks ($\Delta_{\text{gap}} \approx 0.34$, RASR@100% = 12.27) concentrate fragility in a small hub set, so hub protection gives the highest return. Random networks ($\Delta_{\text{gap}} \approx 0.11$, RASR@100% = 28.07) spread fragility evenly, yielding less benefit. Use the degree heterogeneity ratio $k_{\max}/\langle k \rangle$ as a diagnostic: values $\gg 1$ confirm hub dominance (Fig. 2). The large σ values for hub-dominated networks further underscore this: a single hub's fate introduces variance of 12–18 RASR units, meaning hub protection is not only the highest-return strategy but also the most effective variance-reduction intervention.

6.5 *Connectivity Collapse in Sparse Networks*

A WS-equivalent network at $N = 2{,}695$, $\langle k \rangle \approx 2.0$ gives HDA ANC = 0.017 and RF ANC = 0.022 ($\Delta_{\text{gap}} = 0.005$). Near the percolation threshold any removal collapses the network regardless of strategy. Hub protection does not help here — only increasing connectivity does.

6.6 *Limitations and Future Work*

The current study has several limitations. First, BCnns via exhaustive path enumeration limits HBnnsAGP to $N \leqslant 200$; larger networks require faster approximations such as random-walk betweenness estimators. Second, RASR is evaluated with uniform i.i.d. ASR per node; in real systems failure probabilities are correlated (e.g. nodes sharing the same subnet or geographic zone), and degree-dependent heterogeneous ASR profiles remain unexplored. Third, results are limited to unweighted undirected

For Review Only

graphs; directed or weighted topologies, common in transport and power networks, may exhibit different fragility orderings. Fourth, the BA and WS model parameters ($m = 2$, $k = 4$, $p = 0.3$) were fixed; the sensitivity of the fragility ordering to these parameters — especially the WS rewiring probability and the BA attachment number — has not been systematically studied. Fifth, extending PRQMC to larger scales requires cluster-level parallelism beyond `multiprocessing.Pool`. Future work should address these gaps and investigate formal convergence bounds for RASR estimates at intermediate ASR levels.

7. Conclusion

Network topology type is an independent, structurally stable determinant of robustness under probabilistic attack. A cross-topology analysis using PRQMC and HBnnsAGP across four network models, five ASR scenarios, and sizes from $N = 34$ to $N = 10{,}670$ yields four main results:

- The fragility ordering Scale-Free $<$ Real-World $<$ Small-World $<$ Random is invariant to ASR level, strategy, and network size — rooted in structural properties visible before any simulation.
- Random networks are $2.29\times$ more robust than scale-free ones, confirmed by both ANC and RASR independently.
- Halving ASR raises RASR by 42–44 units regardless of topology: structural weakness cannot be offset by reducing attack reliability.
- Near the percolation threshold, targeted and random attacks become indistinguishable.

Practitioners should compute $k_{\max}/\langle k \rangle$: a heavy-tailed distribution flags a network where protecting a small hub set delivers most of the robustness gain and also reduces the largest source of outcome variance. Future work will extend PRQMC to larger scales and investigate degree-dependent ASR profiles.

Funding

This work received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data Availability

The Python code implementing PRQMC, HBnnsAGP, and BCnns, along with all experimental scripts, is available at `[https://github.com/Akshay000000/Networks.git]`. The Zachary Karate Club dataset is publicly available via the NetworkX library (`networkx.karate_club_graph()`).

Acknowledgement

The authors acknowledge the foundational RASR framework of [6] upon which this cross-topology analysis is built. All experiments used the open-source NetworkX library and `scipy.stats.qmc`. The Karate Club dataset was introduced by Zachary [11]. AI writing assistance tools were used during manuscript preparation and are disclosed in accordance with journal policy.

References

For Review Only

1. Barabási, A.-L. (2013) Network science. *Philos. Trans. Roy. Soc. A*, **371**, 20120375.
2. Al Musawi, A. F., Roy, S. & Ghosh, P. (2023) Examining indicators of complex network vulnerability across diverse attack scenarios. *Scientific Reports*, **13**, 18208.
3. Albert, R., Jeong, H. & Barabási, A.-L. (2000) Error and attack tolerance of complex networks. *Nature*, **406**, 378–382.
4. Eiselt, H. (2018) Destabilization of terrorist networks. *Chaos Solitons Fractals*, **108**, 111–118.
5. Pastor-Satorras, R., Castellano, C., Van Mieghem, P. & Vespignani, A. (2015) Epidemic processes in complex networks. *Rev. Mod. Phys.*, **87**, 925.
6. Yang, F. & Wang, Y. (2023) Analyzing the robustness of complex networks with attack success rate. *Entropy*, **25**, 1508.
7. Schneider, C. M., Moreira, A. A., Andrade Jr., J. S., Havlin, S. & Herrmann, H. J. (2011) Mitigation of malicious attacks on networks. *Proc. Natl. Acad. Sci. USA*, **108**, 3838–3841.
8. Artime, O., Grassia, M., De Domenico, M. *et al.* (2024) Robustness and resilience of complex networks. *Nature Reviews Physics*, **6**, 114–131.
9. Barabási, A.-L. & Albert, R. (1999) Emergence of scaling in random networks. *Science*, **286**, 509–512.
10. Watts, D. J. & Strogatz, S. H. (1998) Collective dynamics of small-world networks. *Nature*, **393**, 440–442.
11. Zachary, W. W. (1977) An information flow model for conflict and fission in small groups. *J. Anthropol. Res.*, **33**, 452–473.
12. Newman, M. E. J. (2004) Fast algorithm for detecting community structure in networks. *Phys. Rev. E*, **69**, 066133.
13. Iyer, S., Killingback, T., Sundaram, B. & Wang, Z. (2013) Attack robustness and centrality of complex networks. *PLOS ONE*, **8**, e59613.
14. Fan, C., Zeng, L., Sun, Y. & Liu, Y.-Y. (2020) Finding key players in complex networks through deep reinforcement learning. *Nat. Mach. Intell.*, **2**, 317–324.
15. Callaway, D. S., Newman, M. E. J., Strogatz, S. H. & Watts, D. J. (2000) Network robustness and fragility: percolation on random graphs. *Phys. Rev. Lett.*, **85**, 5468–5471.
16. Cohen, R., Erez, K., ben-Avraham, D. & Havlin, S. (2001) Breakdown of the internet under intentional attack. *Phys. Rev. Lett.*, **86**, 3682–3685.

For Review Only